

TEST CLOSURE REPORT

Banking Transaction Management System (BTMS)

1. Project Information

Project Name	Banking Transaction Management System (BTMS)
Testing Type	Database Testing
Prepared By	Testing Team
Environment	Windows 10, MySQL
Test Start Date	11-05-2026
Test End Date	13-05-2026

2. Scope of Testing

The following database modules were included in the testing scope:

- Customers
- Accounts
- Transactions
- Transaction Categories
- Fees
- Branches
- Employees

Validated Areas

- Primary Key constraints
- Foreign Key constraints
- CHECK constraints
- ENUM value validation
- Trigger execution logic
- Automatic balance updates
- Fee calculation logic
- Daily transfer limit validation
- Duplicate record prevention
- Database business rules enforcement

Out of Scope

- Front-End User Interface (UI)
- APIs and Integration Services
- Reporting Dashboards
- DELETE operations

3. Test Execution Summary

Test Scenario	Total TC	Passed	Failed	Blocked
Account Management	12	9	3	0
Customer Management	12	10	2	0
Transaction Categories	8	6	2	0
Fees & Charges	12	7	5	0
Branches & Employees	10	9	1	1
Transaction Management	15	10	4	1

Overall Testing Statistics

Total Test Cases	69
Passed Test Cases	51
Failed Test Cases	17
Blocked Test Cases	1
Pass Percentage	73.9%

4. Defect Summary

Severity Level	Count
Critical	1
High	7
Medium	4
Low	3

Key Defects Identified

- Daily transfer limit validation failed under specific edge-case scenarios.
- Trigger execution produced inconsistent behavior during rapid concurrent updates.
- Phone number field lacked strict numeric validation.
- Certain transaction validation scenarios were not handled correctly during bulk operations.

5. Business Rule Validation Status

Business Rule	Status
Email uniqueness enforced	Passed
Balance cannot be negative	Passed
Amount greater than zero validation	Passed
Foreign key relationships enforced	Passed
ENUM values validated	Passed
Automatic balance updates	Failed
Failed transactions do not affect balance	Passed
Daily transfer limits enforced	Failed (Edge Case)
Fee limit $\leq 10\%$ enforced	Passed
Receiver account nullable for deposits	Passed

6. Constraints & Trigger Validation

Successfully Validated

- Primary key auto-increment functionality
- Foreign key enforcement
- CHECK constraint validation
- ENUM restriction validation
- Automatic fee record insertion
- Automatic balance update triggers

Issues Identified

- Trigger logic requires improved handling during concurrent or rapid transactions.
- Missing validation for non-numeric phone number values.
- Edge cases in transfer-limit validation require additional handling.

7. Entry & Exit Criteria

Entry Criteria

- Business and database requirements finalized
- Database build deployed successfully to QA environment
- Required test data prepared and validated

Exit Criteria

- All critical defects resolved
- High-severity defects reviewed and accepted
- Test execution completed
- Test summary and defect reports generated
- System approved for next testing phase

8. Risks & Outstanding Issues

- Trigger logic may cause inconsistent results under high transaction concurrency.
- Daily transfer limit validation requires stronger boundary-condition handling.
- Phone number field requires regex or format validation.
- Performance and stress testing under heavy transaction load were not included in this testing cycle.

9. Lessons Learned

1. Trigger validation requires extensive edge-case and concurrency testing.
2. Business rules should be validated at both database and application levels.
3. Proper naming conventions for constraints and triggers simplify debugging and maintenance.
4. Early validation of input fields reduces downstream database inconsistencies.
5. Concurrent transaction scenarios should be included in future testing phases.

10. Recommendations

- Enhance transfer-limit validation logic for edge-case handling.
- Optimize trigger performance for concurrent transactions.
- Implement strict phone number format validation.
- Conduct performance, load, and stress testing.
- Add additional negative and boundary test scenarios for transaction processing.

11. Final Conclusion

Database testing for the Banking Transaction Management System (BTMS) was completed successfully within the planned testing period.

The database demonstrated strong relational integrity, reliable constraint enforcement, and stable trigger execution across most tested scenarios. Core business rules related to account management, transaction processing, fee handling, and balance validation operated as expected.

However, several issues were identified, primarily related to daily transfer limit edge cases, trigger behavior during concurrent updates, and missing phone number validation.

Although these defects do not critically impact the core database structure, they should be resolved before production release.

Based on the executed test scope and current results, the system is considered sufficiently stable to proceed to the next phase of integration and application-level testing after addressing the reported defects.